

# D<sup>3</sup>: Dispute-Driven Debate for Stabilizing Multi-Agent Reasoning in Large Language Models

Yifan Liu<sup>1</sup>, Shiqi Sun<sup>2</sup>, Yu Wang<sup>3</sup>, Shuaishuai Zhang<sup>1</sup>, and Jingyuan Li<sup>1\*</sup>

<sup>1</sup> School of Computer and Artificial Intelligence, Beijing Technology and Business University, Beijing, China

2431062123@st.btbu.edu.cn, 2431062108@st.btbu.edu.cn,  
li.jingyuan.jerry@btbu.edu.cn

<sup>2</sup> State Key Laboratory of AI Safety, Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China

sunshiqi24b@ict.ac.cn

<sup>3</sup> Zhengzhou University, Zhengzhou, China

wangyuuu@gs.zzu.edu.cn

**Abstract.** Single-agent reasoning approaches, including chain-of-thought prompting, typically exhibit limited capacity for self-correction, making it difficult to revise erroneous reasoning once a conclusion has been formed. Multi-Agent Debate (MAD) has emerged as a promising approach to address this limitation by leveraging adversarial interaction between agents. However, we identify a prevalent yet underexplored failure mode: spinning-in-place, where multi-round debate devolves into repetitive argumentation without substantive reasoning progress. Our analysis reveals that this stagnation stems from the lack of explicit mechanisms for identifying and resolving key disagreements. To address this limitation, we propose Dispute-Driven Debate (D<sup>3</sup>), a reasoning framework that restructures multi-round debate dynamics around explicit disputes by explicitly modeling and resolving salient disagreements. D<sup>3</sup> dynamically identifies key disagreements, addresses them through focused sub-debates, and integrates the resulting conclusions back into the main reasoning trajectory. Experiments on six reasoning benchmarks show that D<sup>3</sup> consistently suppresses spinning-in-place behavior, improves the effectiveness of dispute resolution, and achieves competitive accuracy with substantially fewer redundant interactions.

**Keywords:** Multi-Agent Debate · Dispute-Driven Reasoning · Spinning-in-Place · Reasoning Stabilization · Large Language Models.

## 1 Introduction

Large language models (LLMs) have demonstrated strong capabilities on general language tasks [10], yet complex reasoning remains challenging [7].

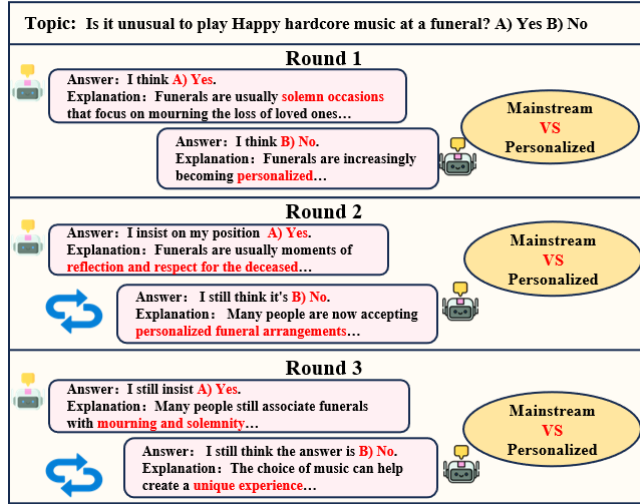
---

\* Corresponding author: li.jingyuan.jerry@btbu.edu.cn

To address this, researchers have explored human-like problem-solving strategies such as chain-of-thought prompting [18] and self-reflection [15], which iteratively refine solutions through self-generated feedback. However, these single-agent methods are prone to Degeneration-of-Thought (DoT) [13]: once a model establishes confidence in its initial reasoning path, it struggles to generate novel perspectives through reflection alone, even when its stance is incorrect.

Multi-Agent Debate (MAD) [1,14,20] has emerged as a promising alternative, leveraging adversarial interaction between agents holding opposing positions to stimulate divergent thinking and mitigate the rigidity of single-agent reasoning. Ideally, iterative exchange of opposing arguments should progressively refine the reasoning process. However, in practice, we observe that multi-round interaction does not reliably translate into sustained reasoning progress. Instead, debate dynamics often stagnate: agents repeatedly revisit the same unresolved disagreement across rounds without introducing new evidence or reasoning paths—a failure mode we term *spinning-in-place* (Fig. 1).

Further analysis reveals that this stagnation arises not from insufficient interaction, but from the absence of explicit mechanisms for identifying and resolving key disagreements. When the locus of disagreement is not precisely identified, agents tend to reinforce their existing positions or converge superficially, rather than making targeted progress on the actual point of contention. Motivated by this observation, we propose Dispute-Driven Debate (D<sup>3</sup>), a framework that explicitly detects salient disputes during debate, resolves them through focused sub-debates, and integrates the outcomes back into the main reasoning trajectory—transforming multi-agent debate from unconstrained interaction into a structured process guided by dispute resolution.



**Fig. 1.** An Illustrative Example of the “Spinning-in-Place” Phenomenon in Multi-Agent Debate

Our main contributions are summarized as follows:

- We identify and formalize a key failure mode in multi-agent LLM debate: *spinning-in-place*, showing how multi-round interaction can devolve into repetitive discussion and inefficient token consumption when key disagreements are not effectively identified or advanced. This analysis reveals intrinsic limitations of existing methods from a **process-level perspective**.
- We propose **Dispute-Driven Debate (D<sup>3</sup>)**, a structured multi-agent reasoning framework centered on the evolution of disputes. By explicitly regulating the **emergence, advancement, and absorption of disputes**, D<sup>3</sup> transforms multi-agent debate from unconstrained interaction into a reasoning process guided by dispute resolution.
- Experiments on six representative reasoning benchmarks with two widely used LLMs show that D<sup>3</sup> **outperforms standard multi-agent debate on many tasks while substantially reducing redundant interaction**, achieving a more robust trade-off between reasoning quality and computational efficiency.

## 2 Multi-Agent Debate Spinning-in-Place

This section identifies spinning-in-place as an underexplored failure mode of multi-agent debate (MAD), where reasoning stagnates instead of progressively refining, thereby motivating the Dispute-Driven Debate (D<sup>3</sup>) framework.

### 2.1 Problem Setup

Given a test query  $Q$  (optionally with context  $C$ ), two agents—Affirmative and Negative—engage in a fixed-round debate. Each agent initializes an answer with its rationale and iteratively revises its position after observing the opponent’s arguments. A Judge then produces the final answer based on the debate process.

We evaluate multi-agent LLM debate on six publicly available benchmark datasets.

- **Counterintuitive AR** [13]: A counterintuitive arithmetic reasoning benchmark where surface-level heuristics often fail.
- **GSM8k** [4]: A math word-problem benchmark for multi-step arithmetic reasoning.
- **MATH** [9]: A broad mathematical reasoning benchmark across multiple domains.
- **MMLU** [8]: A general-knowledge benchmark covering 57 subjects across STEM, humanities, and social sciences.
- **StrategyQA** [6]: A yes/no commonsense reasoning benchmark requiring multi-hop inference.
- **HumanEval** [3]: A code generation benchmark assessing the ability to solve programming problems from natural-language prompts.

**Table 1.** Prevalence of Multi-Round Debates and Spinning-in-Place Behavior, with Spinning-in-Place Rates Computed over Multi-Round Instances

| Dataset                    | Single-Agent Accuracy | MAD Accuracy | Rounds | Multi-round (%) | Spinning-in-Place (%) |
|----------------------------|-----------------------|--------------|--------|-----------------|-----------------------|
| GPT-4o-mini vs GPT-4o-mini |                       |              |        |                 |                       |
| CIAR                       | 0.51                  | 0.54         | 2.900  | 70.00           | 75.26                 |
| GSM8K                      | 0.86                  | 0.92         | 2.610  | 58.91           | 64.17                 |
| MATH                       | 0.66                  | 0.72         | 2.918  | 69.35           | 61.97                 |
| MMLU                       | 0.72                  | 0.81         | 3.879  | 92.08           | 77.95                 |
| StrategyQA                 | 0.51                  | 0.54         | 4.560  | 98.38           | 84.80                 |
| HumanEval                  | 0.64                  | 0.72         | 2.744  | 100.00          | 65.73                 |

All experiments are conducted with GPT-4o-mini at a fixed decoding temperature of 0.7. Final answers are semantically normalized to eliminate superficial variations. We report overall accuracy, average termination rounds, and the Spinning-in-Place Rate, which captures reasoning stagnation when semantically similar outputs fail to advance the core dispute.

## 2.2 Phenomenon Overview

While multi-agent debate aims to refine reasoning through iterative adversarial exchange, our experiments reveal that many debates stagnate, with agents repeatedly producing semantically similar responses that fail to resolve the core dispute despite additional rounds.

Spinning-in-Place is a process-level failure mode in multi-round debates, characterized by:

- Both sides exhibit minimal semantic change across consecutive rounds.
- Core points of disagreement remain unresolved over extended interactions. rounds introduce no new evidence, reasoning paths, or position updates.

From a reasoning dynamics perspective, Spinning-in-Place denotes stalled evolution, where multi-round interactions produce highly overlapping semantic trajectories that repeatedly reuse existing arguments, incurring increasing computational cost with diminishing reasoning returns.

**2.3 Quantitative Evidence of Spinning-in-Place** Table 1 summarizes behavioral statistics of multi-round MAD across datasets, revealing three consistent patterns.

- **Multi-round debates are prevalent**, with over 60% of instances requiring multiple rounds to terminate, suggesting that debate often extends beyond a single exchange.
- **Spinning-in-Place becomes prevalent in multi-round debates**, occurring in over 60% of cases across tasks and approaching 90% in some, indicating that extended debate often yields limited reasoning progress.

- **Simply increasing debate rounds fails to mitigate reasoning stagnation**; performance improves only when new and correct reasoning signals are introduced.
- **Spinning-in-Place causes significant reasoning inefficiency**, incurring high cost with limited information gain across tasks.

Overall, performance degradation in MAD stems from the absence of effective dispute-advancement mechanisms rather than insufficient discussion, causing additional rounds to repeat existing reasoning trajectories.

### 3 Our Framework: Dispute-Driven Debate

Building on these findings, we propose Dispute-Driven Debate (D<sup>3</sup>), a mainline-local coupled framework that improves multi-round reasoning via dynamic dispute detection, localized resolution, and feedback-guided mainline progression (Fig. 2).

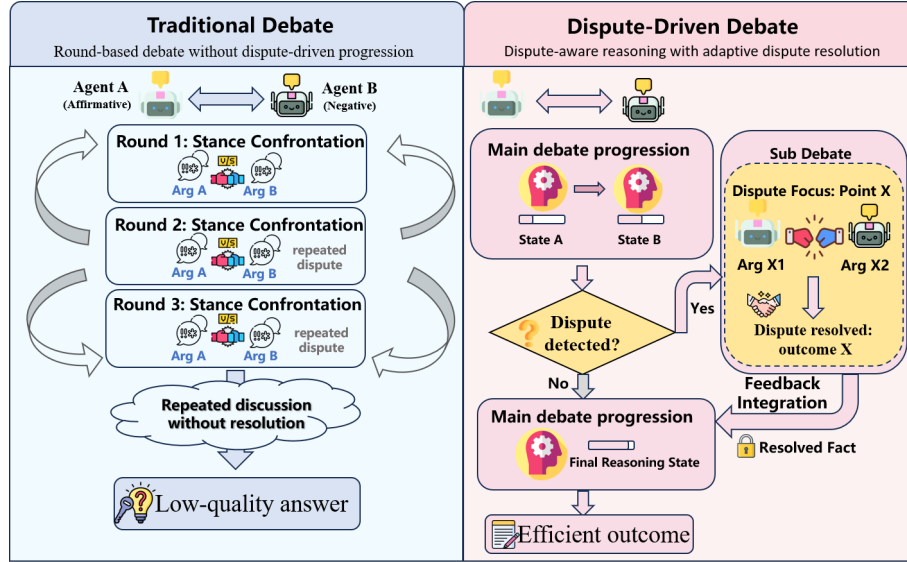


Fig. 2. Traditional Multi-Agent Debate vs. Dispute-Driven Debate

#### 3.1 Mainline Reasoning

The mainline debate is the core reasoning backbone of D<sup>3</sup>, enabling multi-round adversarial reasoning and a cumulative reasoning chain. It consists of two debating agents: the Affirmative (👤) and the Negative (👤), representing the supporting and opposing stances, respectively. The agents alternately contribute

arguments on the central question to advance reasoning. Let  $H^{(r-1)}$  denote the historical context up to round  $r-1$ ; the output of agent  $i$  at round  $r$  is formalized as:

$$h_i^{(r)} = D_i(H^{(r-1)}) \quad (1)$$

where  $D_i(\cdot)$  denotes the generation function of debater  $i$ , and  $h_i^{(r)}$  denotes the output of agent  $i$  at round  $r$ . The updated history is defined as:

$$H^{(r)} = H^{(r-1)} \cup \{h_1^{(r)}, h_2^{(r)}\} \quad (2)$$

In the mainline debate, affirmative and negative stances are prompt-anchored, while arguments evolve dynamically with dialogue history. Through round-based interaction, this module expands the reasoning space around the core question, serving as the global semantic backbone of the D<sup>3</sup> framework.

The mainline debate is dynamically scheduled, with salient disputes triggering focused sub-debates whose outcomes guide subsequent mainline reasoning while preserving focus on the core question.

### 3.2 Dispute Detection & Sub-Debate

In multi-agent multi-round reasoning, disagreements are heterogeneous, ranging from explicit stance conflicts to implicit logical or semantic inconsistencies. Single-signal detection therefore fails to reliably capture such nuances, leading to false positives and missed disputes.

We introduce the Dispute Fusion Score (DFS), which detects salient disagreements by fusing three complementary signals—prompt-based, NLI-based, and semantic gap—capturing conflict at the generation, entailment, and semantic levels.

**Prompt-based Signal** ( $s_p^{(r)}$ ). We first employ an LLM to explicitly assess the degree of disagreement between the two sides. Let  $A^{(r)}$  and  $B^{(r)}$  denote the affirmative and negative arguments at round  $r$ , respectively. Given a dedicated prompt, the LLM determines whether a viewpoint conflict exists and outputs a continuous score  $s_p^{(r)} \in [0, 1]$ , reflecting its perceived intensity of disagreement.



#### Prompt for Dispute Detection

You are a debate dispute detector. Based solely on the two given statements from the Affirmative side and the Negative side, determine whether there is a significant dispute between them.

This signal identifies salient semantic stance conflicts and extracts the underlying dispute focus to guide subsequent sub-debates.

**NLI-based signal ( $s_n^{(r)}$ ).** To mitigate linguistic ambiguity in prompt-based judgments, we introduce an auxiliary NLI-based signal  $s_n^{(r)}$ , which leverages a pretrained model to estimate the contradiction probability  $P_{\text{contradict}}$  and the entailment probability  $P_{\text{entail}}$  between the affirmative and negative statements. The resulting semantic conflict intensity is then defined as follows:

$$s_n^{(r)} = P_{\text{contradict}} - P_{\text{entail}} \quad (3)$$

After normalization, the signal lies in  $[0, 1]$ , with higher values indicating stronger semantic contradiction. The signal  $s_n^{(r)}$  quantifies logical compatibility and entailment between statements.

**Semantic Gap signal ( $s_s^{(r)}$ ).** To further quantify differences in a continuous semantic space, we encode  $A^{(r)}$  and  $B^{(r)}$  into vector representations  $\mathbf{v}_a$  and  $\mathbf{v}_b$ , compute their cosine similarity, and define the semantic gap signal as follows:

$$s_{\text{sem}}^{(r)} = 1 - \cos(\mathbf{v}_a, \mathbf{v}_b) \quad (4)$$

It measures the continuous semantic gap between two statements in embedding space, complementing discrete decision signals with fine-grained semantic divergence.

Together, these signals capture disputes across explicit judgment, logical consistency, and semantic divergence, enabling multidimensional conflict detection. The Dispute Fusion Score (DFS) is defined as follows:

$$\text{DFS} = \alpha \cdot s_p + \beta \cdot s_n + \gamma \cdot s_s \quad (5)$$

Here,  $\alpha$ ,  $\beta$ , and  $\gamma$  are tunable weights that balance the contributions of different signals. When the DFS exceeds a predefined threshold  $\tau \in [0, 1]$ , the current round is identified as having a significant dispute, triggering the sub-debate module.

$$\text{Trigger}^{(r)} = I[s^{(r)} > \tau] \quad (6)$$

Given the computed dispute score, the system produces a structured four-tuple:

$$\text{DisputeTuple} = \left( \text{DFS}^{(r)}, \text{Focus}, \text{stance}_{\text{aff}}, \text{stance}_{\text{neg}} \right) \quad (7)$$

The tuple contains the dispute score, focus, and initial stances, which define the sub-debate. Once triggered, the sub-debate centers on the dispute focus, initializing debaters from  $\text{basis}_{\text{pos}}$  and  $\text{basis}_{\text{neg}}$  as seed claims.

Through iterative interactions, it aims to surface key arguments, reasoning chains, or potential reconciliation points, after which a sub-judge outputs a localized conclusion:

$$\mathcal{C}^{(r)} = f_{\text{sub}}(h_1^{(r)}, h_2^{(r)}, \text{Focus}) \quad (8)$$

where  $\mathcal{C}^{(r)}$  denotes the sub-debate conclusion. The dispute focus may emerge during reasoning rather than being a predefined subtask. Sub-debates isolate fine-grained disagreements for independent deliberation, preserving mainline coherence while improving reasoning focus.

### 3.3 Feedback Integration

To establish a disagreement–deepening–revision reasoning loop,  $D^3$  dynamically integrates the conclusion of each sub-debate into the mainline history upon completion, helping maintain focus and refine subsequent reasoning trajectories. This process can be formalized as:

$$H^{(r)} \leftarrow H^{(r)} \cup C^{(r)} \quad (9)$$

where  $C^{(r)}$  denotes the conclusion of the  $r$ -th sub-debate, and the updated mainline history  $H^{(r)}$  is used as the context for the next debate round.

We introduce two flexible feedback mechanisms to handle different sub-debate outcomes: **(A) Winner Injection**, which injects the dominant stance into the mainline when one side clearly prevails, correcting prior reasoning biases; **(B) Consensus Synthesis**, which integrates the affirmative and negative viewpoints into a balanced perspective when both are well supported or partially aligned, and feeds it back into the mainline.

This feedback mechanism integrates fine-grained sub-debate signals into the mainline, improving coherence and focus while enabling post-conflict self-correction.

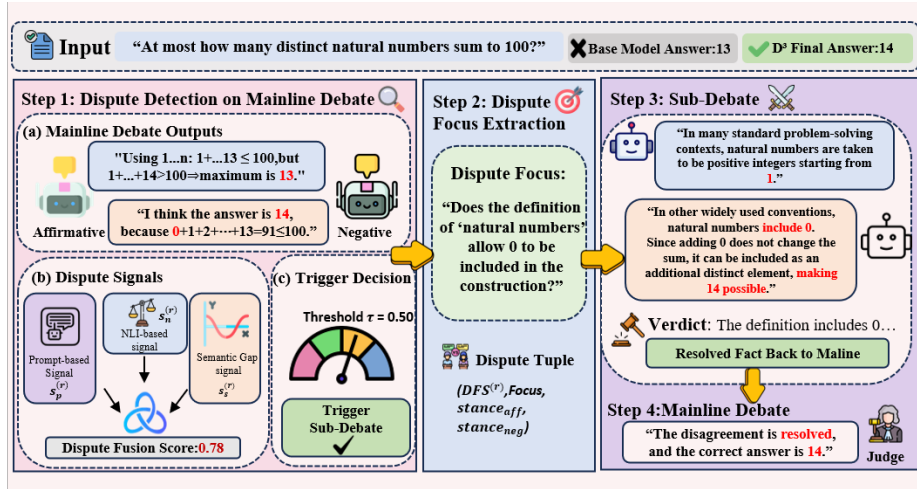


Fig. 3. Overview of the Dispute-Driven Debate ( $D^3$ ) Framework

In summary,  $D^3$  enables structured dispute modeling and adaptive regulation through a closed-loop process of dispute detection (Fig. 3), focused deepening, and systematic absorption.

## 4 Experiment

### 4.1 Experimental Setup

**Table 2.** Benchmark configurations.

| Benchmark           | Category               | Metric           |
|---------------------|------------------------|------------------|
| Counterintuitive AR | Mathematical Reasoning | accuracy, 0-shot |
| GSM8k               | Mathematical Reasoning | accuracy, 0-shot |
| MATH                | Mathematical Reasoning | accuracy, 0-shot |
| MMLU                | General Knowledge      | accuracy, 0-shot |
| StrategyQA          | General Knowledge      | accuracy, 0-shot |
| HumanEval           | Programming            | Pass@1, 0-shot   |

**Tasks & Datasets** We use the same six benchmark datasets as in our preliminary analysis of spinning-in-place behavior in multi-agent LLM debates. Basic statistics are reported in Table 2.

**Baselines** We compare our D<sup>3</sup> (Dispute-Driven Debate) framework against the following baselines:

- **Chain-of-Thought [18] (CoT):** Explicitly generates intermediate reasoning steps prior to the final answer, enhancing performance on multi-step reasoning tasks.
- **Self-Reflect [16] (SR):** Iteratively critiques and revises its own outputs until convergence or a satisfactory solution is reached.
- **Tree-of-Thought [11] (ToT):** Structures reasoning as a tree by exploring multiple sub-questions, enabling branching deliberation and broader reasoning coverage.
- **Multi-Agent Debate [13] (MAD):** Engages two agents with opposing positions in fixed-round debates, with a judge agent aggregating arguments to produce the final decision.

**Agents and Conditions** To assess the generality of Dispute-Driven Debate (D<sup>3</sup>) under controlled and reproducible settings, we evaluate it using GPT-4o-mini and Llama 3 Instruct-7B as backbone models. In each setting, all agents are instantiated from the same base model with identical decoding configurations, ensuring fair comparison. For dispute detection, we incorporate an external discriminative signal to improve reliability. Specifically, we use a pretrained NLI model (`roberta-large-mnli`) to capture entailment and contradiction between opposing arguments, and integrate its outputs into the aggregated dispute score.

We adopt task-specific weight allocations for the three dispute signals: for mathematical reasoning,  $\alpha = 0.4$ ,  $\beta = 0.2$ ,  $\gamma = 0.4$ ; for general knowledge,  $\alpha = 0.4$ ,  $\beta = 0.4$ ,  $\gamma = 0.2$ ; and for programming tasks,  $\alpha = 0.3$ ,  $\beta = 0.1$ ,  $\gamma = 0.6$ .

## 4.2 Evaluation Metric

We adopt two complementary metric categories: outcome-oriented and process-oriented. Outcome-oriented metrics assess final answer quality, while process-oriented metrics characterize multi-round reasoning behavior, including focus

maintenance, error correction, and dispute resolution. As outcome metrics are defined earlier, we focus here on process-oriented metrics.

To characterize the dynamic behavior of multi-round reasoning, we introduce the following two process-oriented metrics:

1. **Spinning-in-Place Rate (SPR).** Given the debater outputs  $h_r$  and  $h_{r+1}$  from consecutive rounds of the mainline debate, we deem a *spinning-in-place* event to occur when the cosine similarity between their semantic representations exceeds a threshold  $\theta$  (e.g.,  $\theta = 0.9$ ).

$$\text{Spin}_r = \begin{cases} 1, & \cos(h_r, h_{r+1}) > \theta \\ 0, & \text{otherwise} \end{cases} \quad (10)$$

The spinning-in-place rate is then defined as:

$$\text{SPR} = \frac{1}{R} \sum_r \text{Spin}_r, \quad (11)$$

where  $R$  denotes the total number of mainline debate rounds.

2. **Dispute Resolution Rate (DRR).** For a dispute  $d_i$  identified during mainline reasoning, subsequent reasoning steps continue its argumentative evolution. If a newly triggered dispute focus  $d_{i+1}$  differs from  $d_i$ , the dispute is considered resolved or shifted; otherwise, repeated re-triggering of  $d_i$  indicates failure to escape the divergence.

$$\text{res}_i = \begin{cases} 1, & d_{i+1} \neq d_i \\ 0, & d_{i+1} = d_i \end{cases} \quad (12)$$

The dispute resolution rate is computed as:

$$\text{DRR} = \frac{1}{N-1} \sum_i \text{res}_i, \quad (13)$$

where  $N$  is the total number of detected disputes. The dispute resolution rate (DRR) quantifies whether the system moves beyond previously identified dispute focuses, avoiding repetitive reasoning cycles.

### 4.3 Main Result

Table 3 shows that D<sup>3</sup> achieves competitive performance across multiple benchmarks. More importantly, these gains arise not from increasing debate rounds, but from a structural change in how disputes are advanced during multi-round reasoning.

D<sup>3</sup> shows clear advantages on tasks involving persistent disagreement and iterative correction, such as counterintuitive arithmetic and multi-step common-sense reasoning. By contrast, simply increasing debate depth often yields limited gains, as standard MAD repeatedly revisits the same disputes without resolution.

**Table 3.** Main Results: Comparison of D<sup>3</sup> with vanilla and advanced multi-agent debate baselines on reasoning benchmarks.

| Baseline              | CIAR        | GSM8K       | MATH        | MMLU        | StrategyQA  | HumanEval   |
|-----------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| <b>GPT-4o-mini</b>    | 51.0        | 86.4        | 66.4        | 72.8        | 51.7        | 64.1        |
| +COT                  | 53.8        | 93.6        | 72.8        | 80.7        | <b>57.5</b> | 70.2        |
| +SR                   | 52.0        | 92.1        | <b>73.9</b> | 82.1        | 56.0        | 66.7        |
| +TOT                  | 48.0        | 89.4        | 71.2        | 76.9        | 53.2        | 68.8        |
| +MAD                  | 54.6        | 92.3        | 72.6        | 81.8        | 54.7        | 72.2        |
| <b>+D<sup>3</sup></b> | <b>56.8</b> | <b>94.2</b> | 72.1        | <b>82.3</b> | 56.2        | <b>74.0</b> |
| <b>Llama 3</b>        | 37.0        | 66.8        | 62.0        | 68.8        | 50.5        | 60.2        |
| +COT                  | 46.0        | <b>74.8</b> | 66.4        | 72.2        | <b>54.6</b> | 64.2        |
| +SR                   | 42.5        | 71.8        | 64.6        | 70.2        | 52.2        | 62.3        |
| +TOT                  | 43.2        | 72.8        | 64.0        | 71.8        | 51.8        | 62.8        |
| +MAD                  | 48.0        | 74.0        | 65.2        | 72.0        | 53.8        | 66.2        |
| <b>+D<sup>3</sup></b> | <b>49.2</b> | 74.2        | <b>68.0</b> | <b>74.8</b> | 54.0        | <b>68.5</b> |

**Table 4.** Process-Oriented Evaluation of Debate Dynamics in MAD and D<sup>3</sup>

| Dataset    | MAD SPR | D <sup>3</sup> SPR↓ | MAD DRR | D <sup>3</sup> DRR↑ |
|------------|---------|---------------------|---------|---------------------|
| CIAR       | 0.75    | 0.42                | 0.15    | 0.34                |
| GSM8K      | 0.64    | 0.56                | 0.28    | 0.38                |
| MATH       | 0.61    | 0.53                | 0.32    | 0.43                |
| MMLU       | 0.77    | 0.60                | 0.18    | 0.27                |
| StrategyQA | 0.84    | 0.47                | 0.23    | 0.35                |
| HumanEval  | 0.65    | 0.48                | 0.21    | 0.36                |

D<sup>3</sup> instead explicitly identifies unresolved disputes and steers subsequent interactions toward their resolution, enabling substantive reasoning progress across rounds.

We also observe that D<sup>3</sup> does not consistently outperform baselines on tasks with limited ambiguity or shallow reasoning. Importantly, its performance does not degrade in these cases, indicating that the added interaction and process-control mechanisms preserve stable reasoning behavior.

Final accuracy alone cannot capture behavioral differences in multi-round reasoning. Table 4 therefore reports process-level metrics, including SPR and DRR.

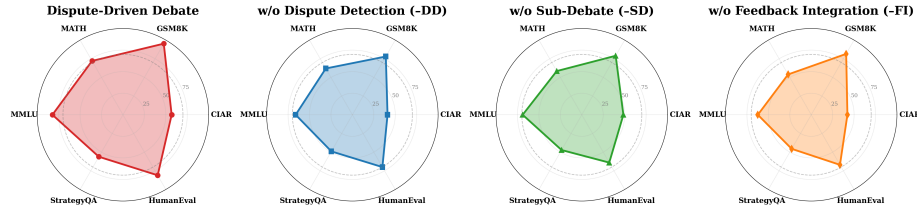
Across all benchmarks, D<sup>3</sup> reduces repetitive debate behavior (lower SPR) and more effectively resolves detected disputes (higher DRR), enabling substantive progress beyond stagnant disagreement points. D<sup>3</sup> derives its gains from structured control over multi-round interactions, enabling additional rounds to contribute new reasoning signals rather than reinforcing stagnant disputes.

By explicitly modeling dispute emergence and resolution,  $D^3$  prevents multi-round debate from degenerating into repetition and instead promotes structured reasoning driven by disagreement resolution.

#### 4.4 Ablation Study

To assess the contribution of individual components within the  $D^3$  (Dispute-Driven Debate) framework, we design three ablated variants:

- **$D^3$  w/o Dispute Detection (–DD):** The main debate no longer explicitly identifies critical disagreements and thus does not trigger sub-topics, proceeding instead via standard pro-con alternation.
- **$D^3$  w/o Sub-Debate (–SD):** Potential disagreements do not interrupt the main debate for localized deliberation; instead, the debate proceeds continuously along the main line.
- **$D^3$  w/o Feedback Integration (–FI):** Sub-debate conclusions are not propagated back to the main debate, depriving it of guidance from localized deliberation.



**Fig. 4.** Component-wise Ablation Analysis of the Dispute-Driven Debate Framework

Our findings (Fig. 4) indicate that, while dispute detection is necessary, the most pronounced and consistent performance degradation arises from removing the sub-debate (–SD) and feedback integration (–FI) mechanisms. This suggests that identifying disagreements alone is insufficient for effective reasoning; rather, progress depends on localized deliberation and the integration of its outcomes into the main debate to enable structured reasoning advancement.

## 5 Analysis

### 5.1 How Does $D^3$ Stabilize Multi-Round Debate?

We quantify the proportion of instances that reach consensus at different debate rounds to characterize the stabilization behavior of multi-round debate (Fig. 5). The results show that  **$D^3$  substantially improves the efficiency of consensus formation in early rounds**: on GPT-4o-mini, approximately 90% of instances reach consensus by Round 3, whereas the standard multi-agent

debate baseline achieves consensus on only about 68% of instances at the same stage. In contrast, the baseline exhibits only a gradual increase in consensus rate across rounds, indicating that additional interactions often fail to translate into stable conclusions. Overall, by enabling early identification and resolution of key disputes, D<sup>3</sup> drives multi-round debate toward a stable state more rapidly, rather than relying on the accumulation of debate rounds.

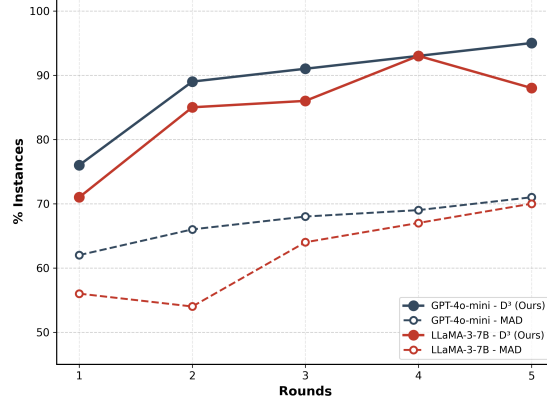


Fig. 5. Stabilizing Multi-Round Debate via Early Consensus Formation

## 5.2 How Cost-Effective Is Dispute-Driven Debate?

From a computational cost perspective, the effectiveness of multi-round debate is determined not by the number of interactions, but by how efficiently the reasoning budget is utilized. As shown in Figure 6, token consumption is decomposed into effective reasoning tokens and spinning-in-place tokens, revealing that vanilla multi-agent debate wastes a substantial portion of tokens on repetitive interactions around unresolved disagreements. In contrast, **D<sup>3</sup> explicitly models dispute emergence and resolution, restructuring the cost distribution by reducing spinning-in-place tokens and reallocating the reasoning budget toward informative rounds.** Even when overall token usage is comparable to MAD, D<sup>3</sup> achieves substantially higher effective reasoning per token, thereby enabling more cost-effective reasoning under a controlled computational budget.

## 5.3 How Sensitive Is D<sup>3</sup> to Weight Configuration?

Table 5 evaluates the sensitivity of D<sup>3</sup> to different weight configurations with  $\gamma$  fixed at 0.2, using MMLU and StrategyQA as representative commonsense reasoning benchmarks. As  $\alpha$  increases and  $\beta$  decreases accordingly, performance generally improves and then stabilizes. D<sup>3</sup> achieves the best accuracy on both

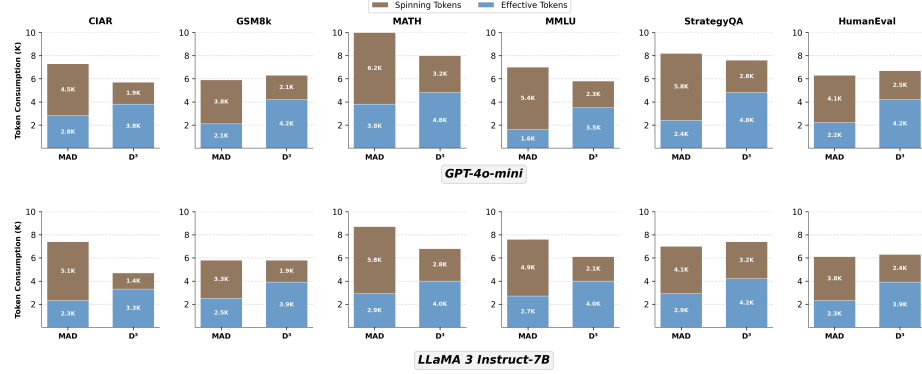


Fig. 6. Reallocating Tokens from Redundant Interaction to Effective Reasoning

Table 5. Robustness to Weight Configuration ( $\gamma$  fixed).

| Task              | $\alpha = 0$<br>$\beta = r$ | $\alpha = 0.25r$<br>$\beta = 0.75r$ | $\alpha = 0.50r$<br>$\beta = 0.50r$ | $\alpha = 0.75r$<br>$\beta = 0.25r$ | $\alpha = r$<br>$\beta = 0$ |
|-------------------|-----------------------------|-------------------------------------|-------------------------------------|-------------------------------------|-----------------------------|
| MMLU (Acc%)       | 80.8                        | 81.7                                | <b>82.3<sup>†</sup></b>             | 82.1                                | 81.4                        |
| StrategyQA (Acc%) | 54.9                        | 55.2                                | <b>56.2<sup>†</sup></b>             | 56.0                                | 55.8                        |

$\gamma$  fixed to  $\gamma_0$ ;  $r = 1 - \gamma_0$ ;  $\beta = r - \alpha$ ;  $\alpha + \beta + \gamma = 1$ .

datasets at  $\alpha = 0.4$ ,  $\beta = 0.4$ , and  $\gamma = 0.2$ . While optimal weights may vary across tasks due to different dependencies on dispute signals, D<sup>3</sup> remains robust to weight variations and supports task-specific tuning for improved performance.

## 6 Related Work

**Multi-Agent Debate for LLM Reasoning** Multi-agent debate has emerged as a promising paradigm for improving LLM reasoning through structured interaction among agents. Du et al.[5] first proposed a multiagent debate framework where multiple LLM instances propose and debate their responses over multiple rounds to reach a common final answer, demonstrating improvements in mathematical reasoning and factual accuracy. Liang et al.[13] introduced a "tit-for-tat" adversarial debate mechanism with a judge to address Degeneration-of-Thought in single-agent reasoning. Subsequent work has explored various directions to enhance debate effectiveness, including sparse communication topologies to reduce redundancy [12], heterogeneous agents with specialized roles [1], and round-table conference structures for consensus building [2]. While these approaches improve debate through better agent configurations or communication structures, they treat disagreement resolution as an implicit byproduct of iterative interaction rather than an explicit objective. When key disagreements are not precisely identified, additional rounds often reinforce existing positions rather than advance the core dispute.

***Disagreement Handling in Multi-Agent Systems*** Existing multi-agent debate frameworks typically resolve disagreements through passive mechanisms such as majority voting [5], judge-based adjudication [13], or natural convergence toward consensus. Recent work has sought to improve these mechanisms by introducing confidence-weighted aggregation [19], learned consensus functions [2], and adaptive termination criteria based on agreement levels [17]. However, these approaches operate at the outcome level—aggregating final answers after debate concludes—rather than intervening during the reasoning process itself. They do not explicitly identify what the agents disagree about or why the disagreement persists.

## 7 Conclusion

We identify a key failure mode of multi-agent debate: unresolved disagreements can lead to repetitive reasoning. To address this, we propose Dispute-Driven Debate (D<sup>3</sup>), which resolves salient disputes through focused sub-debates and feedback integration. Experiments on multiple reasoning benchmarks show that D<sup>3</sup> reduces redundant interaction and improves reasoning efficiency under comparable token budgets. Future work will investigate adaptive dispute modeling and the theory of dispute-driven reasoning.

## 8 Limitations

Although D<sup>3</sup> is evaluated on representative benchmarks, they do not fully capture the diversity of dispute patterns across tasks. Future work will extend evaluation to larger and more diverse datasets. While D<sup>3</sup> improves token efficiency in multi-round debate, it still incurs higher computational cost than single-agent methods, motivating lighter dispute-aware reasoning frameworks.

## Acknowledgments

This work is supported in part by the Henan Provincial Key Research and Development Program of China (No. 241111211900).

## References

1. Chan, C.M., Chen, W., Su, Y., Yu, J., Xue, W., Zhang, S., Fu, J., Liu, Z.: Chateval: Towards better llm-based evaluators through multi-agent debate. In: The Twelfth International Conference on Learning Representations (2024), <https://openreview.net/forum?id=FQepisCUWu>
2. Chen, J., Saha, S., Bansal, M.: Reconcile: Round-table conference improves reasoning via consensus among diverse llms. In: ACL (2024)
3. Chen, M.: Evaluating large language models trained on code. arXiv preprint arXiv:2107.03374 (2021)

4. Cobbe, K., Kosaraju, V., Bavarian, M., Chen, M., Jun, H., Kaiser, L., Plappert, M., Tworek, J., Hilton, J., Nakano, R., et al.: Training verifiers to solve math word problems. arXiv preprint arXiv:2110.14168 (2021)
5. Du, Y., Li, S., Torralba, A., Tenenbaum, J.B., Mordatch, I.: Improving factuality and reasoning in language models through multiagent debate. In: ICML (2024)
6. Geva, M., Khashabi, D., Segal, E., Khot, T., Roth, D., Berant, J.: Did aristotle use a laptop? a question answering benchmark with implicit reasoning strategies. *Transactions of the Association for Computational Linguistics* **9**, 346–361 (2021)
7. Gou, Z., Shao, Z., Gong, Y., Shen, Y., Yang, Y., Duan, N., Chen, W.: Critic: Large language models can self-correct with tool-interactive critiquing. In: ICLR (2024), <https://openreview.net/forum?id=Sx038qxjek>
8. Hendrycks, D., Burns, C., Basart, S., Zou, A., Mazeika, M., Song, D., Steinhardt, J.: Measuring massive multitask language understanding. In: ICLR (2021), <https://openreview.net/forum?id=d7KBjmI3GmQ>
9. Hendrycks, D., Burns, C., Kadavath, S., Arora, A., Basart, S., Tang, E., Song, D., Steinhardt, J.: Measuring mathematical problem solving with the MATH dataset. In: Thirty-fifth Conference on Neural Information Processing Systems Datasets and Benchmarks Track (Round 2) (2021), <https://openreview.net/forum?id=7Bywt2mQsCe>
10. Jiao, W., Wang, W., Huang, J.t., Wang, X., Shi, S., Tu, Z.: Is chatgpt a good translator? yes with gpt-4 as the engine. arXiv preprint arXiv:2301.08745 (2023)
11. Kargupta, P., Agarwal, I., August, T., Han, J.: Tree-of-debate: Multi-persona debate trees elicit critical thinking for scientific comparative analysis. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL 2025) (2025), <https://aclanthology.org/2025.acl-long.1422/>
12. Li, Y., Du, Y., Zhang, J., Hou, L., Grabowski, P., Li, Y., Ie, E.: Improving multi-agent debate with sparse communication topology. In: Findings of EMNLP (2024)
13. Liang, T., He, Z., Jiao, W., Wang, X., Wang, Y., Wang, R., Yang, Y., Shi, S., Tu, Z.: Encouraging divergent thinking in large language models through multi-agent debate. In: Proceedings of the 2024 conference on empirical methods in natural language processing. pp. 17889–17904 (2024)
14. Liu, T., Wang, X., Huang, W., Xu, W., Zeng, Y., Jiang, L., Yang, H., Li, J.: Groupdebate: Enhancing the efficiency of multi-agent debate using group discussion. In: The 25th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2026) (2026), <https://openreview.net/forum?id=KDzl9qlrLi>
15. Madaan, A., Tandon, N., Gupta, P., Hallinan, S., Gao, L., Wiegrefe, S., Alon, U., Dziri, N., Prabhume, S., Yang, Y., et al.: Self-refine: Iterative refinement with self-feedback, 2023. URL <https://arxiv.org/abs/2303.17651> (2023)
16. Shinn, N., Cassano, F., Gopinath, A., Narasimhan, K., Yao, S.: Reflexion: Language agents with verbal reinforcement learning. *Advances in Neural Information Processing Systems* **36**, 8634–8652 (2023)
17. Smit, A.P., Grinsztajn, N., Duckworth, P., Barrett, T.D., Pretorius, A.: Should we be going mad? a look at multi-agent debate strategies for llms. In: ICML (2024)
18. Wei, J., Wang, X., Schuurmans, D., Bosma, M., Xia, F., Chi, E., Le, Q.V., Zhou, D., et al.: Chain-of-thought prompting elicits reasoning in large language models. *Advances in neural information processing systems* **35**, 24824–24837 (2022)
19. Xiong, K., Ding, X., Cao, Y., Liu, T., Qin, B.: Examining inter-consistency of large language models collaboration: An in-depth analysis via debate. In: Findings of EMNLP (2023)
20. Zhang, H., Cui, Z., Wang, X., Zhang, Q., Wang, Z., Wu, D., Hu, S.: If multi-agent debate is the answer, what is the question. arXiv preprint arXiv:2502.08788 (2025)